In [13]:
```
from google.colab import files
uploaded = files.upload()
```

Upload widget is only available when the cell has been executed in the current browser session. Please rerun this cell to enable.

```
Saving Mall_Customers - Mall_Customers.xlsx.csv to Mall_Customers -
Mall_Customers.xlsx.csv
```

In [14]:
```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
%matplotlib inline

df = pd.read_csv('Mall_Customers - Mall_Customers.xlsx.csv')
df.info()
print(df.head())

sns.pairplot(df)

features = df.iloc[:, [3, 4]].values

from sklearn.cluster import KMeans
model = KMeans(n_clusters=5, random_state=42, n_init=10)
model.fit(features)
print(model)

Final = df.iloc[:, [3, 4]].copy()
Final['label'] = model.predict(features)
print(Final.head())

sns.set_style("whitegrid")
sns.FacetGrid(Final, hue="label", height=8) \
    .map(plt.scatter, "Annual Income (k$)", "Spending Score (1-100)") \
    .add_legend()
plt.show()

features_el = df.iloc[:, [2, 3, 4]].values
wcss = []

for i in range(1, 10):
    model = KMeans(n_clusters=i, random_state=42, n_init=10)
    model.fit(features_el)
    wcss.append(model.inertia_)

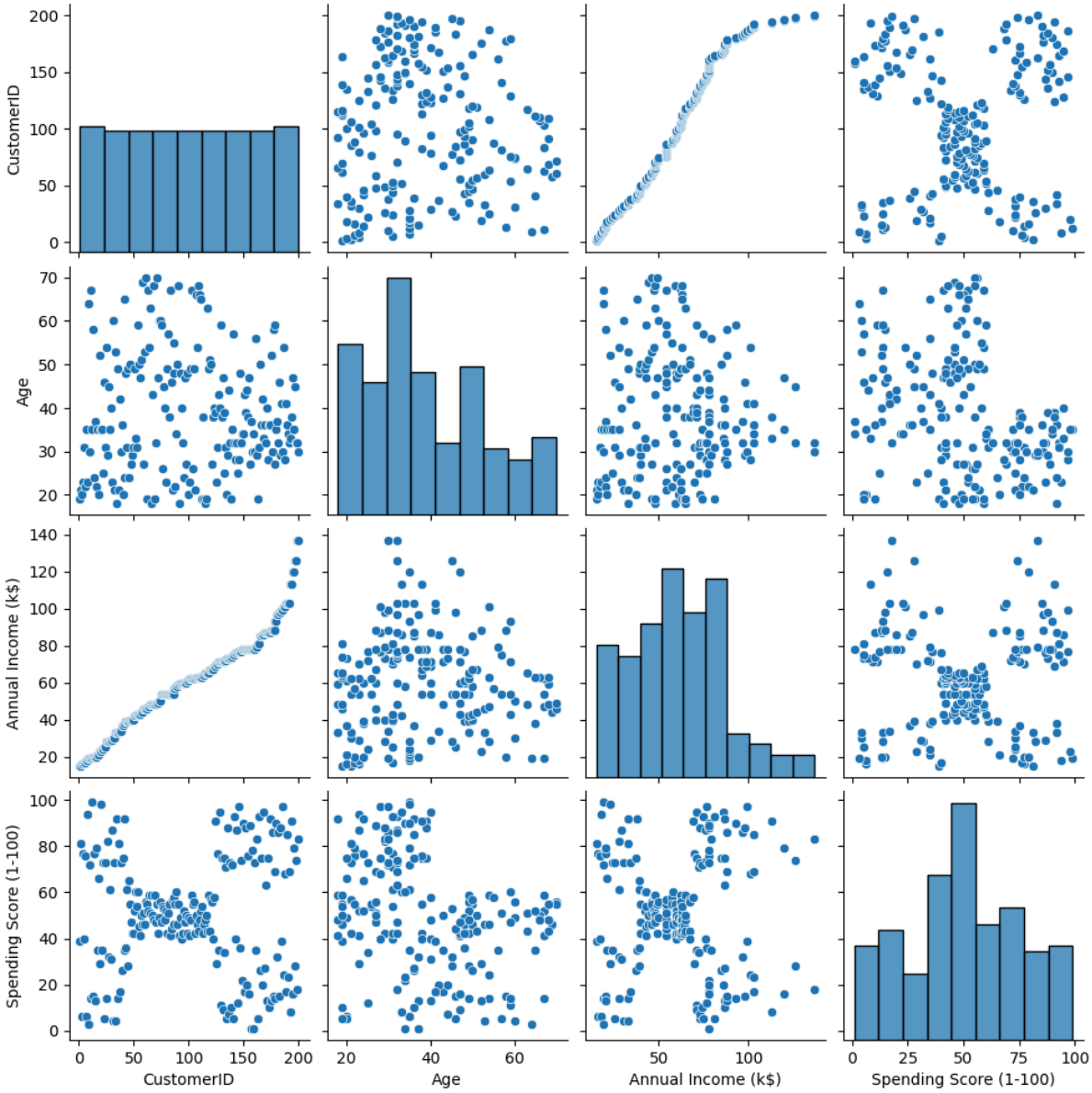
plt.plot(range(1, 10), wcss)
plt.xlabel("Number of Clusters")
plt.ylabel("WCSS")
plt.show()
```

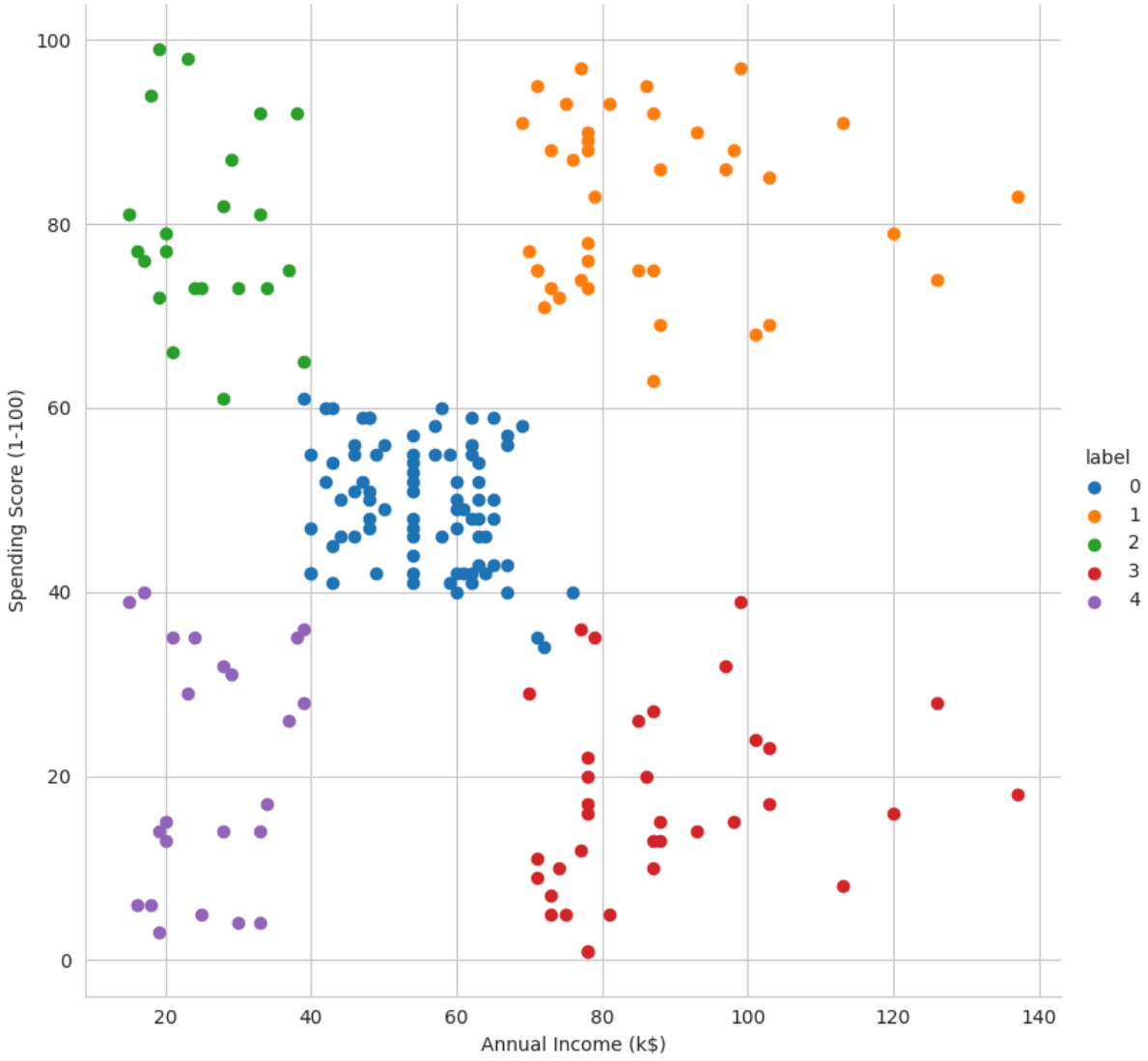
```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 200 entries, 0 to 199
Data columns (total 5 columns):
 #   Column                  Non-Null Count  Dtype
---  ------                  --------------  -----
 0   CustomerID              200 non-null    int64
 1   Gender                  200 non-null    object
 2   Age                     200 non-null    int64
 3   Annual Income (k$)      200 non-null    int64
 4   Spending Score (1-100)  200 non-null    int64
dtypes: int64(4), object(1)
memory usage: 7.9+ KB
```

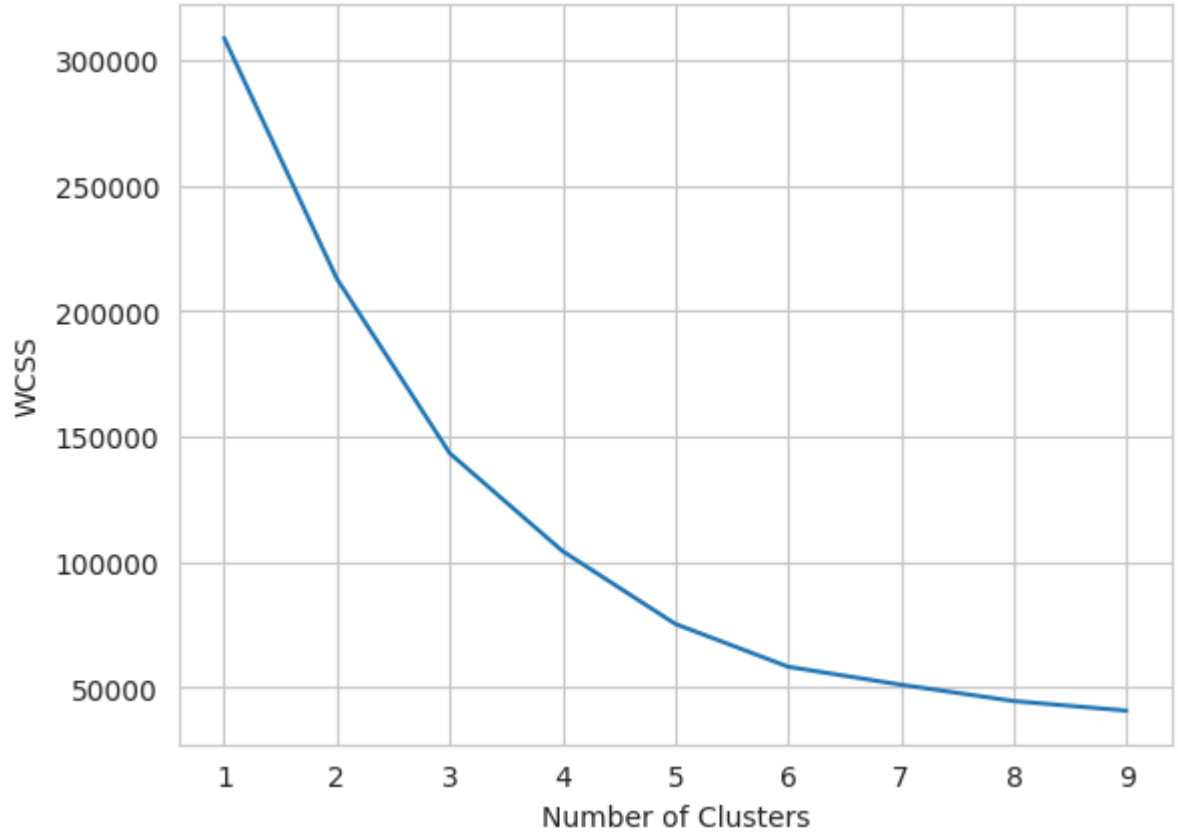
| | CustomerID | Gender | Age | Annual Income (k$) | Spending Score (1-100) |
|---|---|---|---|---|---|
| 0 | 1 | Male | 19 | 15 | 39 |
| 1 | 2 | Male | 21 | 15 | 81 |
| 2 | 3 | Female | 20 | 16 | 6 |
| 3 | 4 | Female | 23 | 16 | 77 |
| 4 | 5 | Female | 31 | 17 | 40 |

```
KMeans(n_clusters=5, n_init=10, random_state=42)
```

| | Annual Income (k$) | Spending Score (1-100) | label |
|---|---|---|---|
| 0 | 15 | 39 | 4 |
| 1 | 15 | 81 | 2 |
| 2 | 16 | 6 | 4 |
| 3 | 16 | 77 | 2 |
| 4 | 17 | 40 | 4 |


CustomerID
Age
Annual Income (k$)
Spending Score (1-100)
CustomerID
Age
Annual Income (k$)
Spending Score (1-100)


Spending Score (1-100)
Annual Income (k$)
label
0
1
2
3
4


WCSS
Number of Clusters